

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping Questions

Course Code:	CSA0303	Course Title:	Data Structures
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Mapping Questions
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
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Section / Batch:	SLOT-C	Date:	11-07-26

Code of Conduct

I SABITHA V 192421415 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V SABITHA

Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'	Concept mapping on classifications of data structures.	1	30
Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.	Linear and Binary search	2	40
Primitive and non-primitive data structure, Linear and non-linear data structure Arrays	Linked List	3	40

Annexure A – Concept mapping questions

- 1) A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.
- 2) A system needs to search for specific records in a large dataset efficiently.]Construct a concept map linking Searching Algorithm → Linear Search → Binary Search → Time Complexity → Efficiency. Include relationships that compare the number of operations and performance of each method. Analyze which algorithm is more efficient and justify your conclusion.
- 3)A library management system stores thousands of book records including title, author, and availability status. Construct a concept map linking Data → Data Structure → Organization → Access Methods → Efficiency. Include different data structures used for storing and retrieving records. Explain how proper organization improves search efficiency and system performance.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts

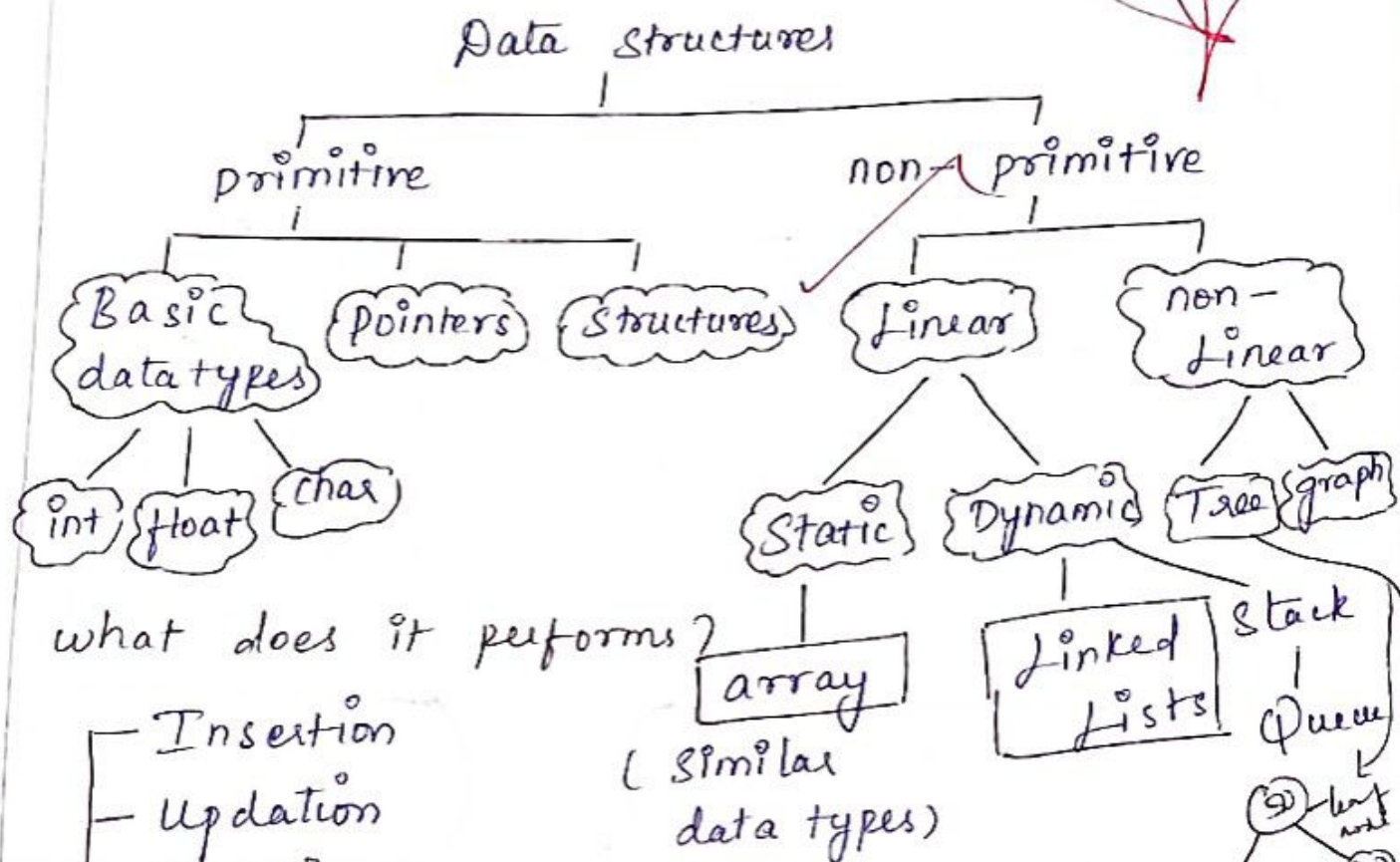
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

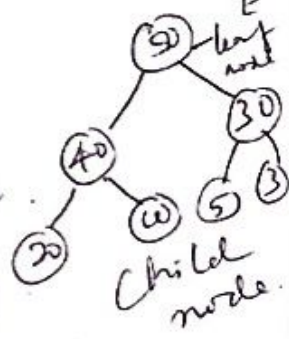
① Data Structures :-

* It is a method of organising or arranging the data in the primary memory so that the users can easily access and stores the data

Classification of data structures



- * Organizes the data.
- * Performs Searching
- * Sorting the data.
- * Easy to store and Access.



For Example :-

array =

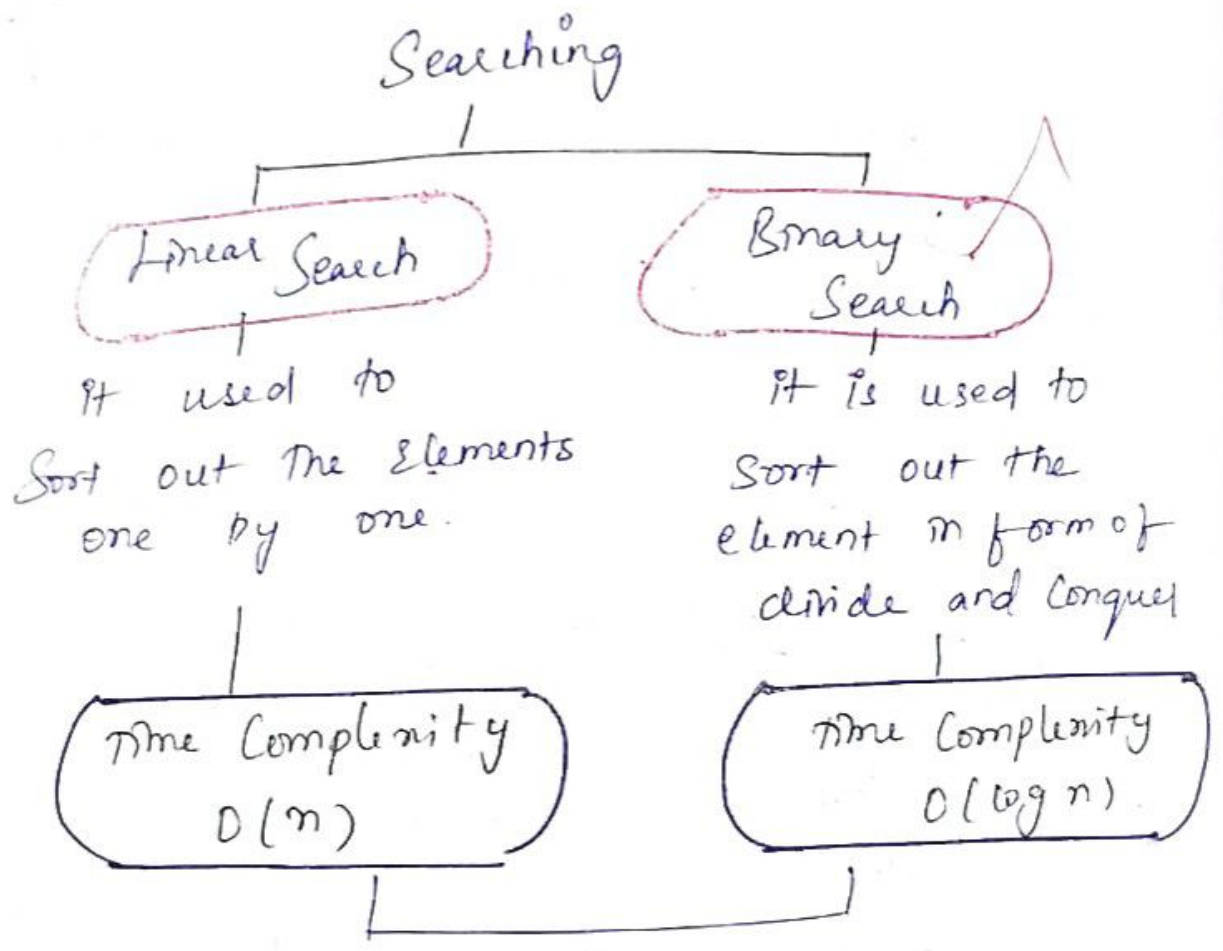
0	1	2	3	4
15	20	70	80	90

Example key

- * It comes under the Linear data structure which performs static form of storing the arrays like no. of vehicles, Speed of vehicles.
- * Efficiently in forms of Tree structures.

Linear Search (Searching):

(2)



Efficiency

- * Compared to Both Binary and Linear search Binary Search is Efficient and faster than Linear Search.

Example for Linear Search program

```
#include <stdio.h>
```

```
int main() {
```

```
    int i, n, key = 30, found = 1;  
    key = 30
```

```
    scanf("%d %d", &n);
```

```
    for (i = 0; i < n; i++)
```

```
        found = 1;
```

```
    if ("Element is found:")
```

```
        break;
```

```
    if ("Element not found:")
```

```
        return 0;
```

```
}
```

} shows slower
than Binary
Search

Binary Search Example to show it faster search

int (i, n)

get an array element of
a[50]

key = 30

found = 1

if (i = 0; i < n; i++) (i+1)

Element found.

if (i == 1)

Element not found

return.

③

Library Management System
Title, Author, Availability
Status.

Library management using data structures

Organisation
using array
shows no. of Books
etc.

Access methods
using Stack,
Queue,
Linked List

Efficiency
Based upon
Time
complexity

Stores the
data in the form
of Arrays.

- Titles
- Authors
- Availability of Books

$A[50] =$

0	1	2	3	4
15	30	70	95	100

{ Key = 30 index = 1
Array shows the
position & index.

Performances in
data structures

- Insert
- Update
- Delete
- Retrieve

15	30	70	90	100
----	----	----	----	-----

insert 50 in place
of 70

① perform sorting

15	30	70	90	100
----	----	----	----	-----

Move one by one

15	30	50	90	100
----	----	----	----	-----

Struct Box

```
{ int bid;  
  char t[20], a[20];
```

```
  struct Book * next  
}
```

head

1	DS	Lia
---	----	-----

2	CS	Arjun
---	----	-------

3	EVS	Prity
---	-----	-------

4	Maths	Shruti
---	-------	--------

4 Books
for management
of Library